```
vyos@vyos:~$ install image
Welcome to the VyOS install program.  This script
will walk you through the process of installing the
VyOS image to a local hard drive.
Would you like to continue? (Yes/No) [Yes]:
Probing drives: OK
Looking for pre-existing RAID groups...none found.
The VyOS image will require a minimum 2000MB root.
Would you like me to try to partition a drive automatically
or would you rather partition it manually with parted?  If
you have already setup your partitions, you may skip this step

Partition (Auto/Parted/Skip) [Auto]:

I found the following drives on your system:
 sda    17179MB


Install the image on? [sda]:

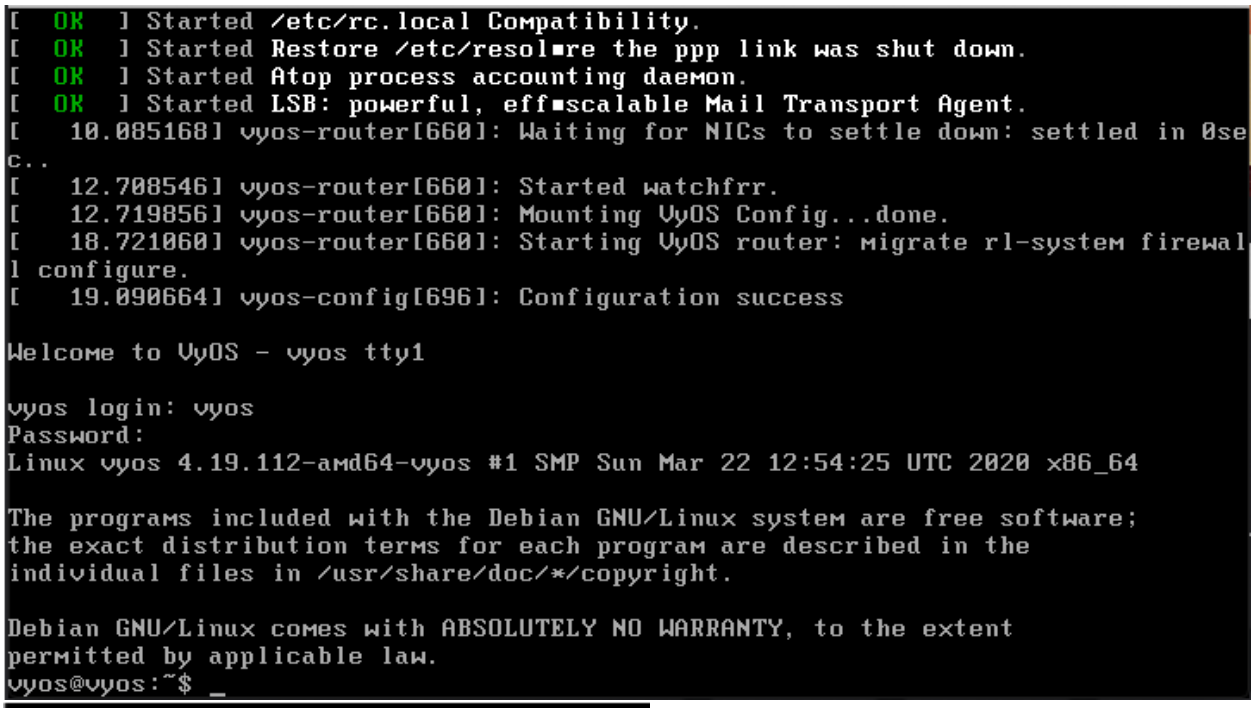
This will destroy all data on /dev/sda.
Continue? (Yes/No) [No]: yes

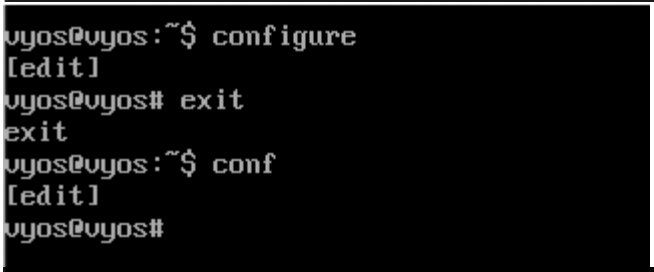
Looking for config files from previous installations on sda1...
How big of a root partition should I create? (2000MB - 17179MB) [17179]MB: _
Mounting /dev/sda1...
What would you like to name this image? [1.3-kroy-202004011525]: OK.  This image
 will be named: 1.3-kroy-202004011525
Copying squashfs image...

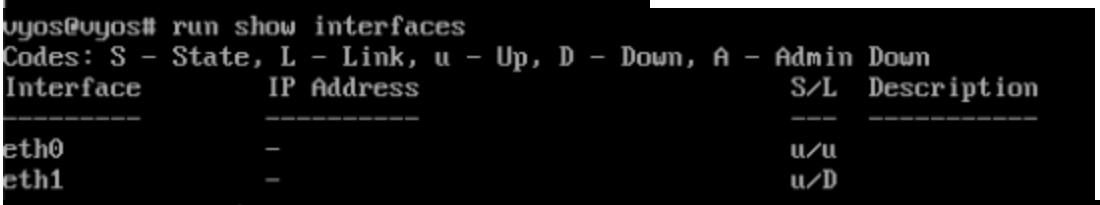
Copying kernel and initrd images...
Done!
I found the following configuration files:
    /opt/vyatta/etc/config/config.boot
    /opt/vyatta/etc/config.boot.default
Which one should I copy to sda? [/opt/vyatta/etc/config/config.boot]:
Copying /opt/vyatta/etc/config/config.boot to sda.
Enter password for administrator account
Enter password for user 'vyos':
Retype password for user 'vyos':
I need to install the GRUB boot loader.
I found the following drives on your system:
 sda    17179MB

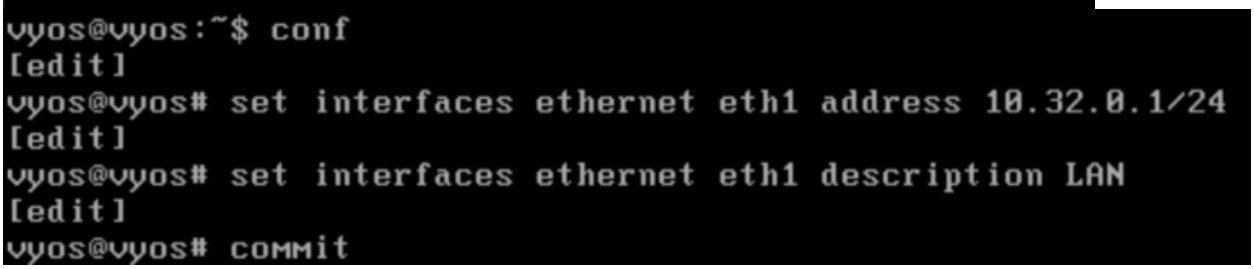

Which drive should GRUB modify the boot partition on? [sda]:

Setting up grub: OK
Done!
vyos@vyos:~$
```


[OK] Started /etc/rc.local Compatibility.
[OK] Started Restore /etc/resol▪re the ppp link was shut down.
[OK] Started Atop process accounting daemon.
[OK] Started LSB: powerful, eff▪scalable Mail Transport Agent.
[10.085168] vyos-router[660]: Waiting for NICs to settle down: settled in 0se
c..
[12.708546] vyos-router[660]: Started watchfrr.
[12.719856] vyos-router[660]: Mounting VyOS Config...done.
[18.721060] vyos-router[660]: Starting VyOS router: migrate rl-system firewal
l configure.
[19.090664] vyos-config[696]: Configuration success
Welcome to VyOS - vyos tty1
vyos login: vyos
Password:
Linux vyos 4.19.112-amd64-vyos #1 SMP Sun Mar 22 12:54:25 UTC 2020 x86_64
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
vyos@vyos:~$ _


vyos@vyos:~$ configure
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ conf
[edit]
vyos@vyos#


vyos@vyos# run show interfaces
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface   IP Address   S/L Description
---------   ----------   --- -----------
eth0   -   u/u
eth1   -   u/D


vyos@vyos:~$ conf
[edit]
vyos@vyos# set interfaces ethernet eth1 address 10.32.0.1/24
[edit]
vyos@vyos# set interfaces ethernet eth1 description LAN
[edit]
vyos@vyos# commit

```
vyos@vyos# edit interfaces ethernet eth1
[edit interfaces ethernet eth1]
vyos@vyos# set address 10.32.0.1/24
[edit interfaces ethernet eth1]
vyos@vyos# set description LAN
[edit interfaces ethernet eth1]
vyos@vyos# commit
```

```
[edit]
vyos@vyos# show interfaces
 ethernet eth0 {
     hw-id 00:50:56:b5:ab:cd
 }
 ethernet eth1 {
     address 10.32.0.1/24
     description LAN
 }
 loopback lo {
 }
[edit]
vyos@vyos# run show interfaces
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface        IP Address                        S/L  Description
---------        ----------                        ---  -----------
eth0             -                                 u/u
eth1             10.32.0.1/24                      u/u  LAN
lo               127.0.0.1/8                       u/u
                 ::1/128
[edit]
vyos@vyos#
```

```
[edit]
vyos@vyos# set service dhcp-server shared-network-name LAN subnet 10.32.0.0/24 range 0 start 10.32.0.50
[edit]
vyos@vyos# set service dhcp-server shared-network-name LAN subnet 10.32.0.0/24 range 0 stop 10.32.0.125
[edit]
vyos@vyos# set service dhcp-server shared-network-name LAN subnet 10.32.0.0/24 range 1 start 10.32.0.200
[edit]
vyos@vyos# set service dhcp-server shared-network-name LAN subnet 10.32.0.0/24 range 1 stop 10.32.0.250
[edit]
vyos@vyos# commit
save
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos#
```

```
[edit]
vyos@vyos# set service dhcp-server shared-network-name LAN subnet 10.32.0.0/24 dns-server 10.32.0.1
[edit]
vyos@vyos# set service dhcp-server shared-network-name LAN subnet 10.32.0.0/24 default-router 10.32.0.1
[edit]
vyos@vyos# commit
save
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos#
```

```
kroy@debian-gui:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens192: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:50:56:[illegible]:d9:65 brd ff:ff:ff:ff:ff:ff
    inet 10.32.0.51/24 brd 10.32.0.255 scope global dynamic noprefixroute ens192
       valid_lft 83998sec preferred_lft 83998sec
    inet6 fe80::a15a:3718:5ff:70c6/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
kroy@debian-gui:~$ ip r
default via 10.32.0.1 dev ens192 proto dhcp metric 100
10.32.0.0/24 dev ens192 proto kernel scope link src 10.32.0.51 metric 100
kroy@debian-gui:~$ cat /etc/resolv.conf
# Generated by NetworkManager
nameserver 10.32.0.1
kroy@debian-gui:~$ ping 10.32.0.1 -c4
PING 10.32.0.1 (10.32.0.1) 56(84) bytes of data.
64 bytes from 10.32.0.1: icmp_seq=1 ttl=64 time=0.489 ms
64 bytes from 10.32.0.1: icmp_seq=2 ttl=64 time=0.584 ms
64 bytes from 10.32.0.1: icmp_seq=3 ttl=64 time=0.157 ms
64 bytes from 10.32.0.1: icmp_seq=4 ttl=64 time=0.645 ms

--- 10.32.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 46ms
rtt min/avg/max/mdev = 0.157/0.468/0.645/0.190 ms
kroy@debian-gui:~$
```

```
kroy@debian-gui:~$ ssh vyos@10.32.0.1
Welcome to VyOS
vyos@10.32.0.1's password:
Linux vyos 4.19.114-amd64-vyos #1 SMP Sat Apr 4 19:51:19 UTC 2020 x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Tue Apr  7 19:01:15 2020 from 10.32.0.51
```

```
vyos@vyos:~$ show configuration
interfaces {
    ethernet eth0 {
        hw-id 00:50:56:b5:ab:cd
    }
    ethernet eth1 {
        address 10.32.0.1/24
        hw-id 00:50:56:b5:87:1a
    }
    loopback lo {
    }
}
service {
    dhcp-server {
        shared-network-name LAN {
            subnet 10.32.0.0/24 {
                default-router 10.32.0.1
                dns-server 10.32.0.1
                range 0 {
                    start 10.32.0.50
                    stop 10.32.0.125
                }
                range 1 {
                    start 10.32.0.200
                    stop 10.32.0.250
                }
            }
        }
    }
    ssh {
        port 22
    }
}
system {
    config-management {
        commit-revisions 100
    }
    console {
        device ttyS0 {
            speed 115200
        }
    }
    host-name vyos
    login {
        user vyos {
            authentication {
                encrypted-password ****************
                plaintext-password ****************
            }
```


system {
config-management {
commit-revisions 100
}
console {
device ttyS0 {
speed 115200
}
}
host-name vyos
login {
user vyos {
authentication {
encrypted-password ****************
plaintext-password ****************
}
}
}
ntp {
server 0.pool.ntp.org {
}
server 1.pool.ntp.org {
}
server 2.pool.ntp.org {
}
}
syslog {
global {
facility all {
level info
}
facility protocols {
level debug
}
}
}
}

```
[edit]
vyos@vyos# set nat source rule 100 source address '10.32.0.0/24'
[edit]
vyos@vyos# set nat source rule 100 outbound-interface eth0
[edit]
vyos@vyos# set nat source rule 100 translation address masquerade
[edit]
vyos@vyos# commit
save
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# show nat source
 rule 100 {
     outbound-interface eth0
     source {
         address 10.32.0.0/24
     }
     translation {
         address masquerade
     }
 }
[edit]
```

```
[edit]
vyos@vyos# set service dns forwarding listen-address 10.32.0.1
[edit]
vyos@vyos# set service dns forwarding allow-from 10.32.0.0/24
[edit]
vyos@vyos# set service dns forwarding cache-size 0
[edit]
vyos@vyos# commit
save
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
```

```
[edit]
vyos@vyos# set service dns forwarding name-server 1.1.1.1
[edit]
vyos@vyos# set service dns forwarding name-server 1.0.0.1
[edit]
vyos@vyos# set service dns forwarding name-server 8.8.8.8
[edit]
vyos@vyos# set service dns forwarding name-server 8.8.4.4
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# show service dns
 forwarding {
     allow-from 10.32.0.0/24
     cache-size 0
     listen-address 10.32.0.1
     name-server 1.1.1.1
     name-server 1.0.0.1
     name-server 8.8.8.8
     name-server 8.8.4.4
 }
[edit]
vyos@vyos#
```

```
vyos@vyos# set system name-server 10.32.0.1
[edit]
vyos@vyos# commit
save
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
```

```
vyos@vyos:~$ conf
[edit]
vyos@vyos# set firewall name LAN-WAN default-action accept
[edit]
vyos@vyos# set firewall name LAN-LOCAL default-action accept
[edit]
vyos@vyos# commit
save
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# set firewall name LOCAL-WAN default-action accept
[edit]
vyos@vyos# set firewall name LOCAL-LAN default-action accept
[edit]
vyos@vyos# commit
save
[edit]
vyos@vyos# save
vyos@vyos# set firewall name WAN-LOCAL default-action drop
[edit]
vyos@vyos# set firewall name WAN-LOCAL rule 5 action accept
[edit]
vyos@vyos# set firewall name WAN-LOCAL rule 5 state established enable
[edit]
vyos@vyos# set firewall name WAN-LOCAL rule 5 state related enable
[edit]
vyos@vyos# set firewall name WAN-LOCAL rule 5 description "Allow EST/Related Traffic"
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# show firewall name WAN-LOCAL
 default-action drop
 rule 5 {
     action accept
     description "Allow EST/Related Traffic"
     state {
         established enable
         related enable
     }
 }
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
```

```
vyos@vyos# set firewall name WAN-LOCAL rule 20 protocol icmp
[edit]
vyos@vyos# set firewall name WAN-LOCAL rule 20 state new enable
[edit]
vyos@vyos# set firewall name WAN-LOCAL rule 20 action accept
[edit]
vyos@vyos# commit
save
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# run show firewall name WAN-LOCAL

----------------------------
Rulesets Information
----------------------------

IPv4 Firewall "WAN-LOCAL":

 Inactive - Not applied to any interfaces or zones.

rule  action   proto     packets  bytes
----  ------   -----     -------  -----
5     accept   all       0        0
  condition - saddr 0.0.0.0/0 daddr 0.0.0.0/0 WAN-LOCAL-5 */

20    accept   icmp      0        0
  condition - saddr 0.0.0.0/0 daddr 0.0.0.0/0 WAN-LOCAL-20 */

10000 drop     all       0        0
  condition - saddr 0.0.0.0/0 daddr 0.0.0.0/0
```

```
[edit]
vyos@vyos# set firewall name WAN-LAN default-action drop
[edit]
vyos@vyos# set firewall name WAN-LAN rule 5 action accept
[edit]
vyos@vyos# set firewall name WAN-LAN rule 5 state established enable
[edit]
vyos@vyos# set firewall name WAN-LAN rule 5 state related enable
[edit]
vyos@vyos# edit firewall name WAN-LAN
[edit firewall name WAN-LAN]
vyos@vyos# set rule 20 protocol icmp
[edit firewall name WAN-LAN]
vyos@vyos# set rule 20 state new enable
[edit firewall name WAN-LAN]
vyos@vyos# set rule 20 action accept
[edit firewall name WAN-LAN]
vyos@vyos# commit
save
[edit firewall name WAN-LAN]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
[edit]
vyos@vyos# set zone-policy zone LAN default-action drop
[edit]
vyos@vyos# set zone-policy zone LAN from WAN firewall name WAN-LAN
[edit]
vyos@vyos# set zone-policy zone LAN from LOCAL firewall name LOCAL-LAN
[edit]
vyos@vyos# set zone-policy zone LAN interface eth1
[edit]
vyos@vyos# set zone-policy zone LOCAL local-zone
[edit]
vyos@vyos# set zone-policy zone LOCAL from LAN firewall name LAN-LOCAL
[edit]
vyos@vyos# set zone-policy zone LOCAL from WAN firewall name WAN-LOCAL
[edit]
vyos@vyos# set zone-policy zone LOCAL default-action drop
[edit]
vyos@vyos# set zone-policy zone WAN from LAN firewall name LAN-WAN
[edit]
vyos@vyos# set zone-policy zone WAN from LOCAL firewall name LOCAL-WAN
[edit]
vyos@vyos# set zone-policy zone WAN interface eth0
[edit]
vyos@vyos# set zone-policy zone WAN default-action drop
[edit]
vyos@vyos# commit
```

```
[edit]
vyos@vyos# delete zone-policy zone
[edit]
vyos@vyos# set zone-policy zone LAN default-action drop
[edit]
vyos@vyos# set zone-policy zone LAN from WAN firewall name WAN-LAN
[edit]
vyos@vyos# set zone-policy zone LAN from LOCAL firewall name LOCAL-LAN
[edit]
vyos@vyos# set zone-policy zone LAN interface eth1
[edit]
vyos@vyos# commit
sa[ zone-policy ]
LOCAL is a from zone under zone LAN
It is either not defined or deleted from config

[[zone-policy]] failed
[[zone-policy zone LAN from WAN]] failed
[[zone-policy zone LAN from LOCAL]] failed
Commit failed
```

```
vyos@vyos# run show zone-policy
--------------------
Name: LAN

Interfaces: eth1

From Zone:
  name                                    firewall
  ----                                    --------
  LOCAL                                   LOCAL-LAN
  WAN                                     WAN-LAN

--------------------
Name: LOCAL

Interfaces: local-zone

From Zone:
  name                                    firewall
  ----                                    --------
  LAN                                     LAN-LOCAL
  WAN                                     WAN-LOCAL

--------------------
Name: WAN
```

```
------------------
Name: WAN

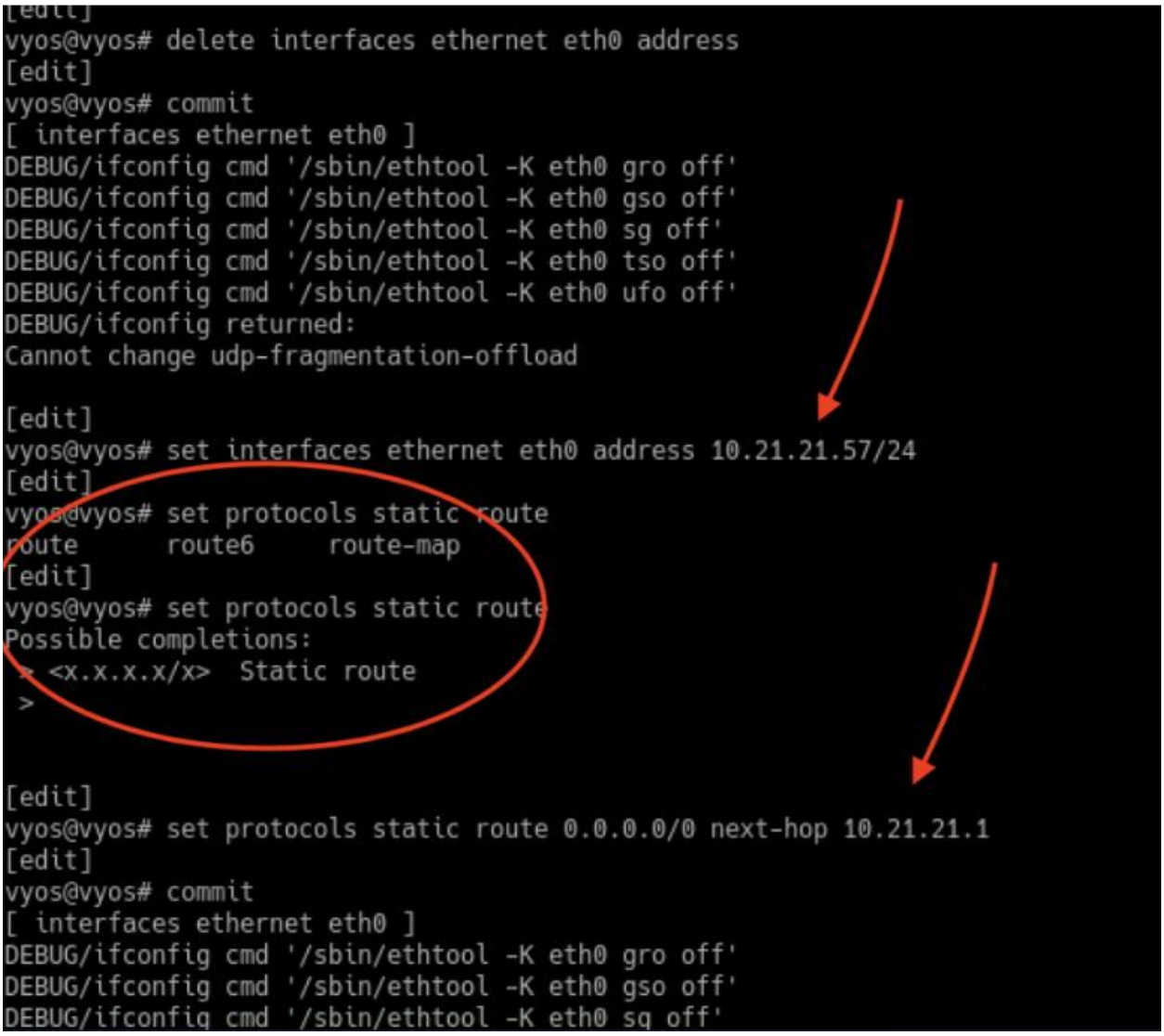
Interfaces: eth0

From Zone:
  name                                    firewall
  ----                                    --------
  LAN                                     LAN-WAN
  LOCAL                                   LOCAL-WAN

[edit]
```

```
[edit]
vyos@vyos# set interfaces ethernet eth0 address dhcp
[edit]
vyos@vyos# set interfaces ethernet eth0 description WAN
[edit]
vyos@vyos# commit
save
[ interfaces ethernet eth0 ]
DEBUG/ifconfig cmd '/sbin/ethtool -K eth0 gro off'
DEBUG/ifconfig cmd '/sbin/ethtool -K eth0 gso off'
DEBUG/ifconfig cmd '/sbin/ethtool -K eth0 sg off'
DEBUG/ifconfig cmd '/sbin/ethtool -K eth0 tso off'
DEBUG/ifconfig cmd '/sbin/ethtool -K eth0 ufo off'
DEBUG/ifconfig returned:
Cannot change udp-fragmentation-offload

[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# run show interfaces
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface        IP Address                        S/L  Description
---------        ----------                        ---  -----------
eth0             10.21.21.57/24                    u/u  WAN
eth1             10.32.0.1/24                      u/u  LAN
lo               127.0.0.1/8                       u/u
                 ::1/128
[edit]
vyos@vyos# run show ip route
```


[edit]
vyos@vyos# delete interfaces ethernet eth0 address
[edit]
vyos@vyos# commit
[interfaces ethernet eth0]
DEBUG/ifconfig cmd '/sbin/ethtool -K eth0 gro off'
DEBUG/ifconfig cmd '/sbin/ethtool -K eth0 gso off'
DEBUG/ifconfig cmd '/sbin/ethtool -K eth0 sg off'
DEBUG/ifconfig cmd '/sbin/ethtool -K eth0 tso off'
DEBUG/ifconfig cmd '/sbin/ethtool -K eth0 ufo off'
DEBUG/ifconfig returned:
Cannot change udp-fragmentation-offload
[edit]
vyos@vyos# set interfaces ethernet eth0 address 10.21.21.57/24
[edit]
vyos@vyos# set protocols static route
route route6 route-map
[edit]
vyos@vyos# set protocols static route
Possible completions:
> <x.x.x.x/x> Static route
>
[edit]
vyos@vyos# set protocols static route 0.0.0.0/0 next-hop 10.21.21.1
[edit]
vyos@vyos# commit
[interfaces ethernet eth0]
DEBUG/ifconfig cmd '/sbin/ethtool -K eth0 gro off'
DEBUG/ifconfig cmd '/sbin/ethtool -K eth0 gso off'
DEBUG/ifconfig cmd '/sbin/ethtool -K eth0 sg off'

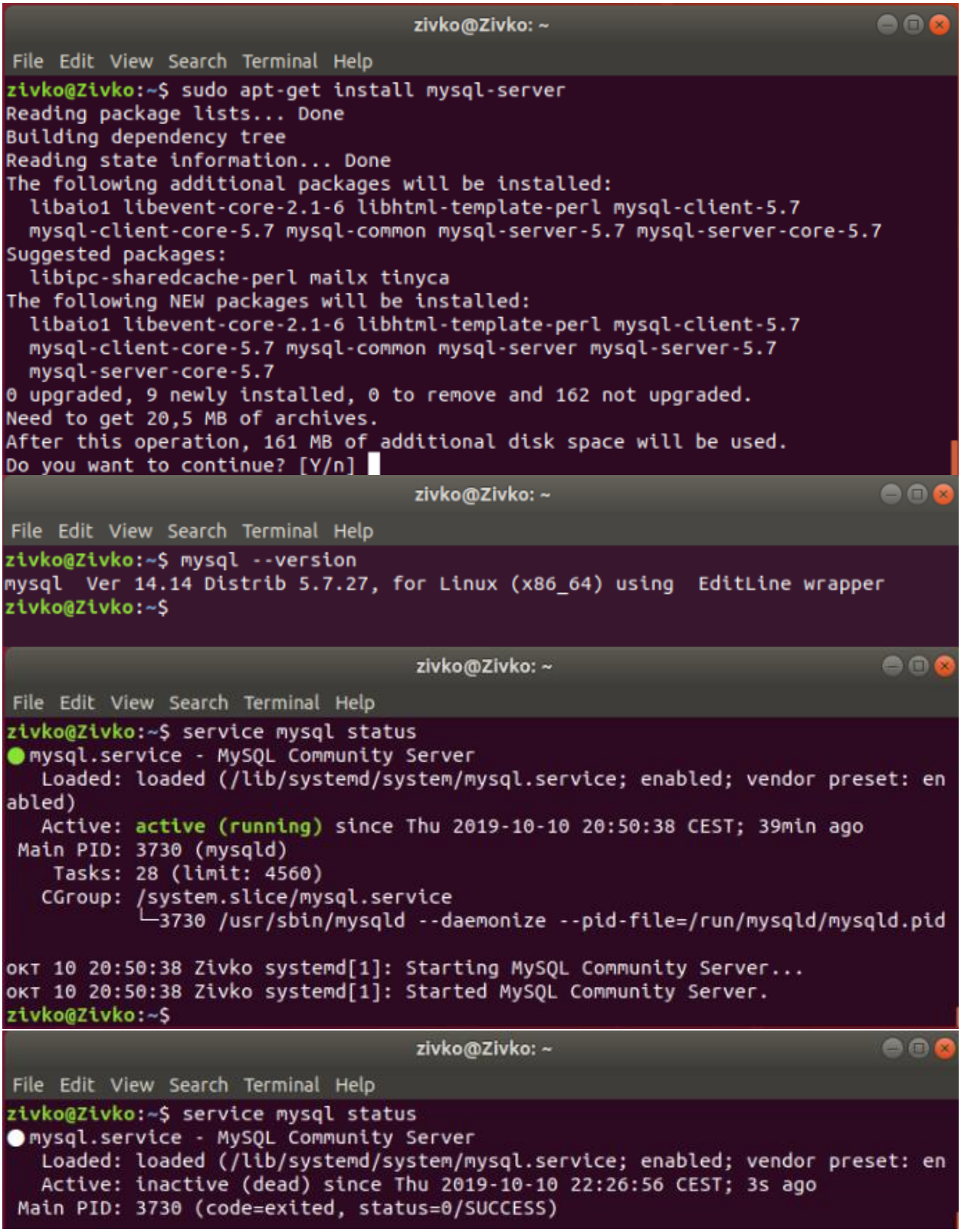
```
[edit]
vyos@vyos# set protocols static route 0.0.0.0/0 next-hop 10.21.21.1
[edit]
vyos@vyos# commit
[ interfaces ethernet eth0 ]
DEBUG/ifconfig cmd '/sbin/ethtool -K eth0 gro off'
DEBUG/ifconfig cmd '/sbin/ethtool -K eth0 gso off'
DEBUG/ifconfig cmd '/sbin/ethtool -K eth0 sg off'
DEBUG/ifconfig cmd '/sbin/ethtool -K eth0 tso off'
DEBUG/ifconfig cmd '/sbin/ethtool -K eth0 ufo off'
DEBUG/ifconfig returned:
Cannot change udp-fragmentation-offload

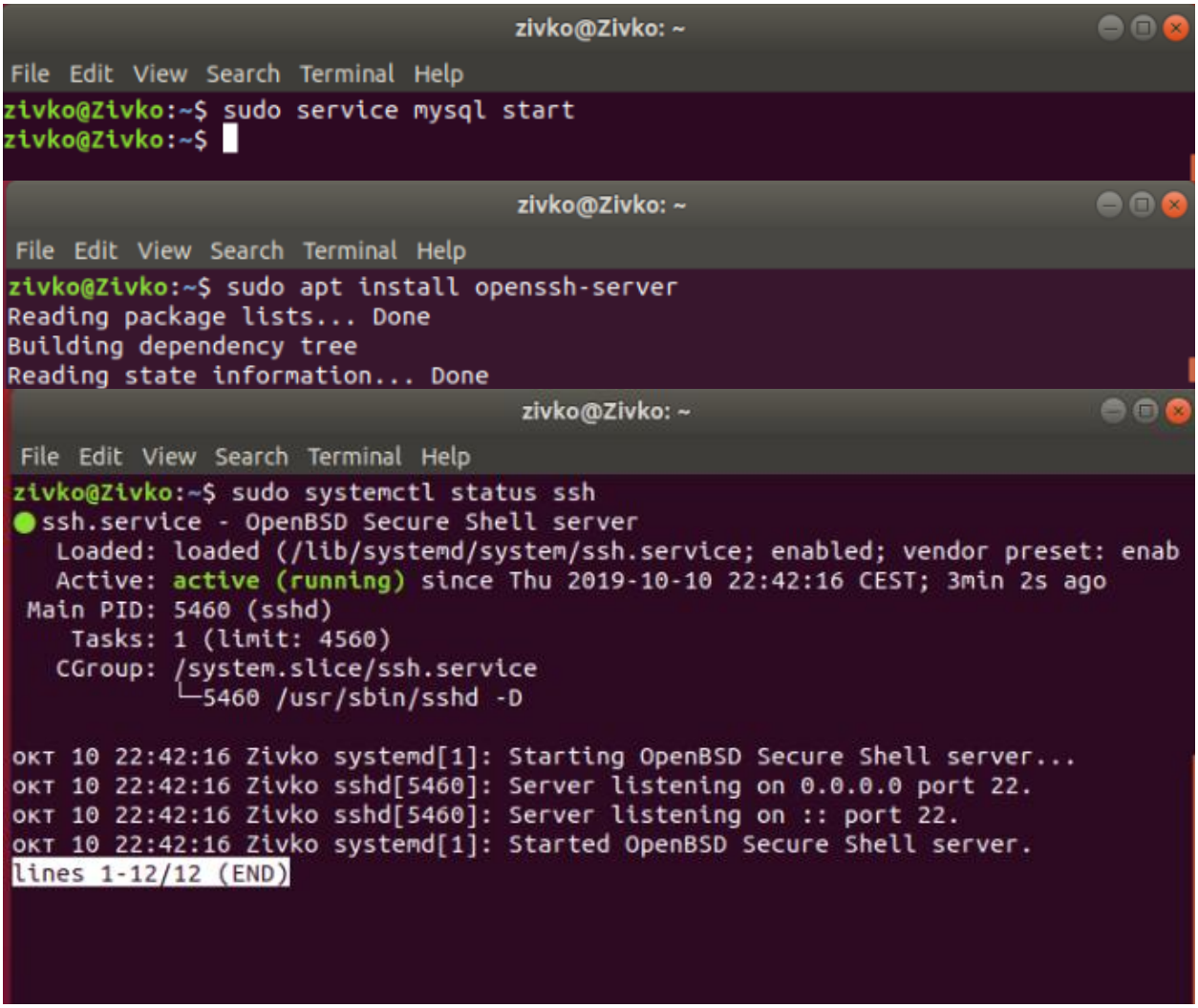
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
```

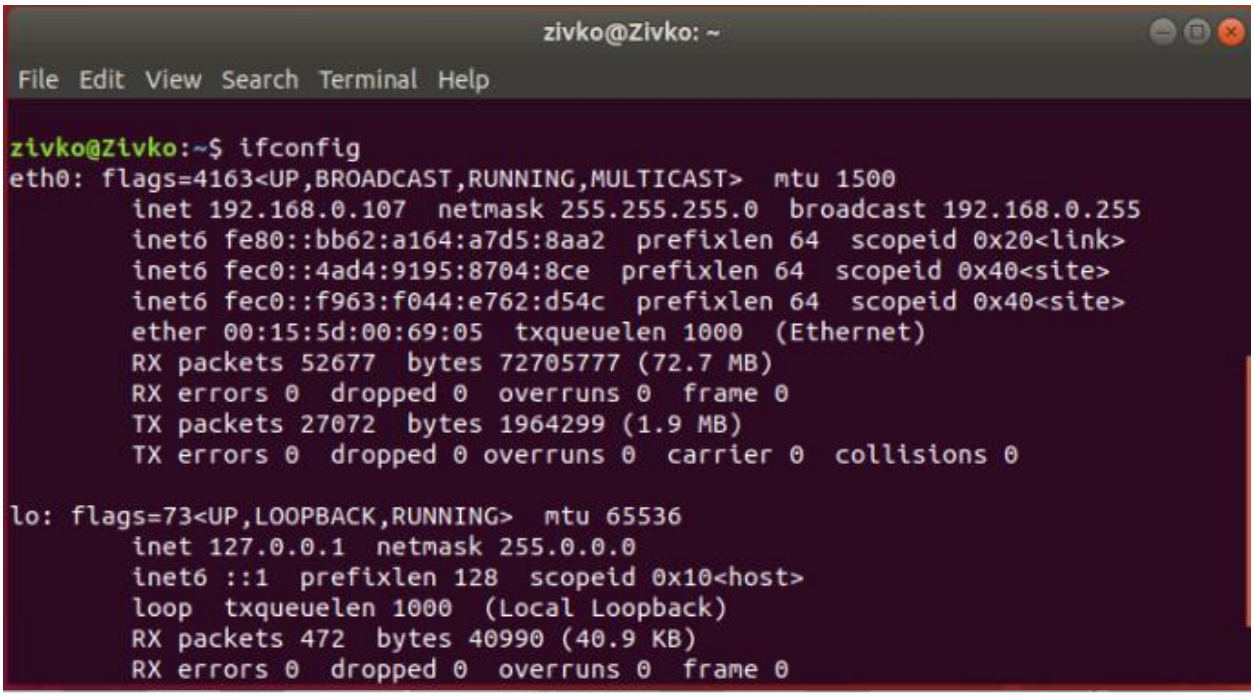
```
[edit]
vyos@vyos# run ping www.google.com count 4
PING www.google.com (216.58.192.196) 56(84) bytes of data.
64 bytes from ord30s25-in-f4.1e100.net (216.58.192.196): icmp_seq=1 ttl=56 time=14.1 ms
64 bytes from ord30s25-in-f4.1e100.net (216.58.192.196): icmp_seq=2 ttl=56 time=14.8 ms
64 bytes from ord30s25-in-f4.1e100.net (216.58.192.196): icmp_seq=3 ttl=56 time=19.3 ms
64 bytes from ord30s25-in-f4.1e100.net (216.58.192.196): icmp_seq=4 ttl=56 time=15.5 ms

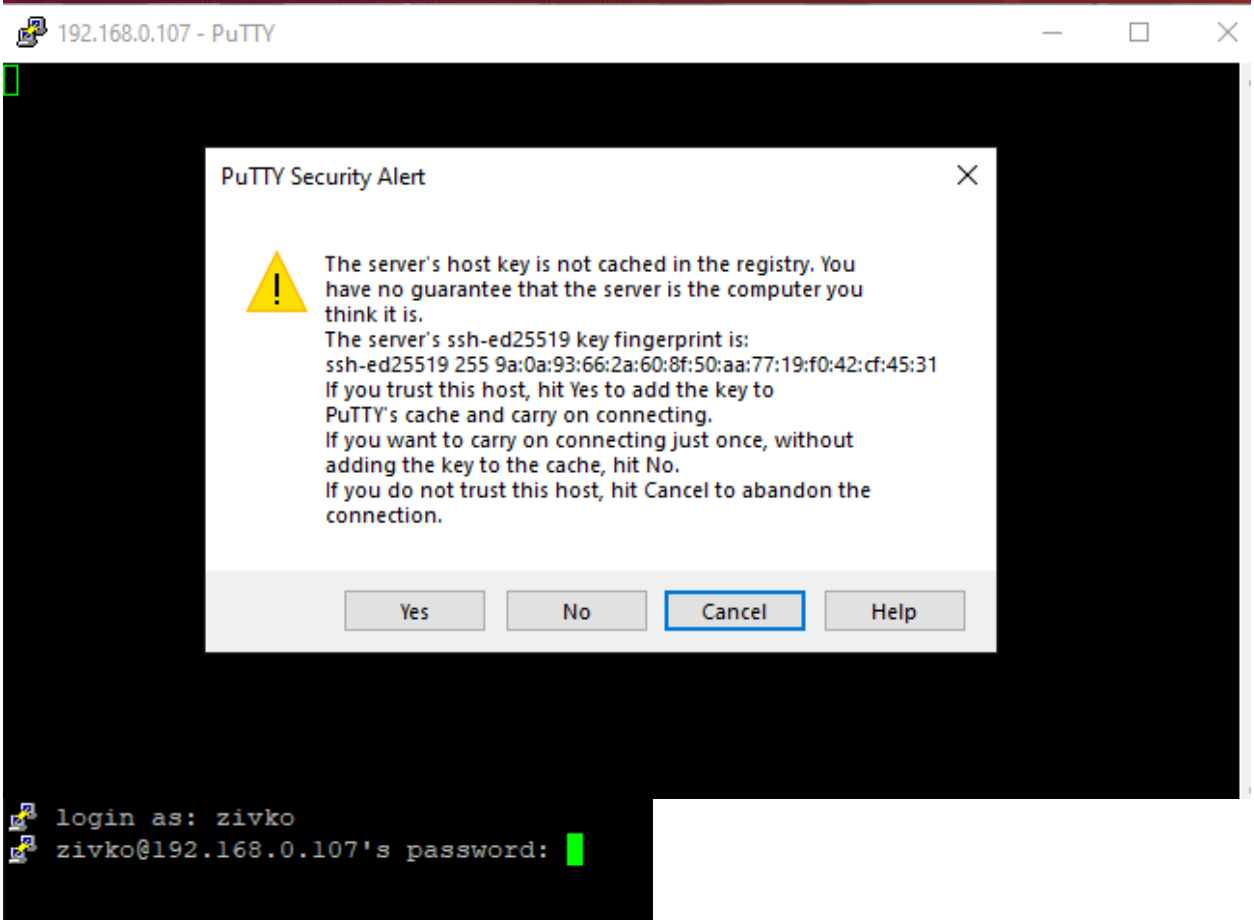
--- www.google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 8ms
rtt min/avg/max/mdev = 14.094/15.927/19.347/2.039 ms
vyos@vyos:~$ exit
logout
Connection to 10.32.0.1 closed.
kroy@debian-gui:~$ ping www.google.com -c 8
PING www.google.com (216.58.192.164) 56(84) bytes of data.
64 bytes from ord36s02-in-f164.1e100.net (216.58.192.164): icmp_seq=1 ttl=55 time=18.7 ms
64 bytes from ord36s02-in-f164.1e100.net (216.58.192.164): icmp_seq=2 ttl=55 time=19.3 ms
64 bytes from ord36s02-in-f164.1e100.net (216.58.192.164): icmp_seq=3 ttl=55 time=18.9 ms
64 bytes from ord36s02-in-f164.1e100.net (216.58.192.164): icmp_seq=4 ttl=55 time=18.8 ms
64 bytes from ord36s02-in-f164.1e100.net (216.58.192.164): icmp_seq=5 ttl=55 time=18.10 ms
64 bytes from ord36s02-in-f164.1e100.net (216.58.192.164): icmp_seq=6 ttl=55 time=18.8 ms
64 bytes from ord36s02-in-f164.1e100.net (216.58.192.164): icmp_seq=7 ttl=55 time=19.2 ms
64 bytes from ord36s02-in-f164.1e100.net (216.58.192.164): icmp_seq=8 ttl=55 time=18.7 ms

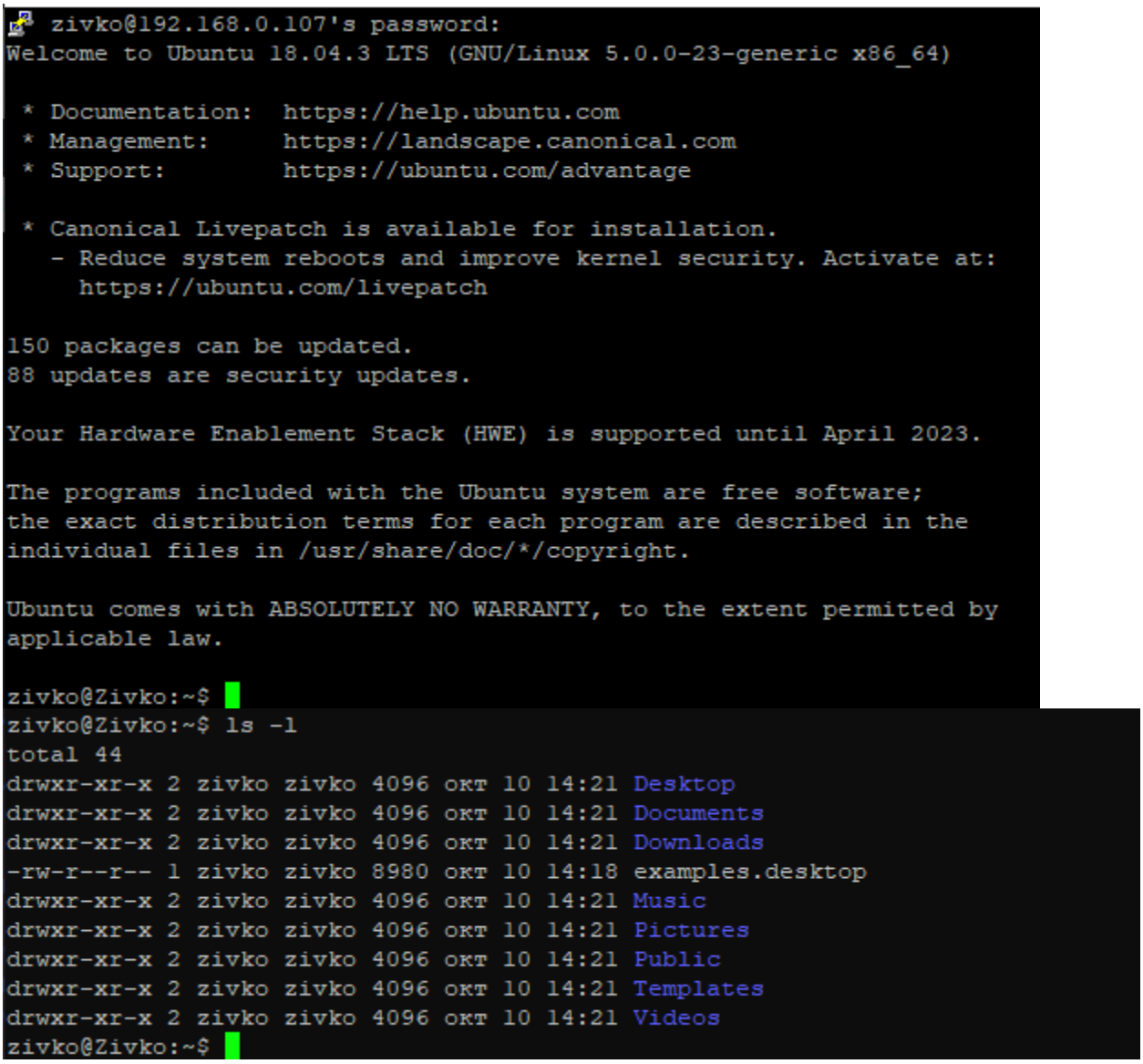
--- www.google.com ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 17ms
rtt min/avg/max/mdev = 18.707/18.934/19.288/0.241 ms
kroy@debian-gui:~$
```


zivko@Zivko: ~
File Edit View Search Terminal Help
zivko@Zivko:~$ sudo apt-get install mysql-server
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
libaio1 libevent-core-2.1-6 libhtml-template-perl mysql-client-5.7
mysql-client-core-5.7 mysql-common mysql-server-5.7 mysql-server-core-5.7
Suggested packages:
libipc-sharedcache-perl mailx tinyca
The following NEW packages will be installed:
libaio1 libevent-core-2.1-6 libhtml-template-perl mysql-client-5.7
mysql-client-core-5.7 mysql-common mysql-server mysql-server-5.7
mysql-server-core-5.7
0 upgraded, 9 newly installed, 0 to remove and 162 not upgraded.
Need to get 20,5 MB of archives.
After this operation, 161 MB of additional disk space will be used.
Do you want to continue? [Y/n]
zivko@Zivko: ~
File Edit View Search Terminal Help
zivko@Zivko:~$ mysql --version
mysql Ver 14.14 Distrib 5.7.27, for Linux (x86_64) using EditLine wrapper
zivko@Zivko:~$
zivko@Zivko: ~
File Edit View Search Terminal Help
zivko@Zivko:~$ service mysql status
mysql.service - MySQL Community Server
Loaded: loaded (/lib/systemd/system/mysql.service; enabled; vendor preset: en
abled)
Active: active (running) since Thu 2019-10-10 20:50:38 CEST; 39min ago
Main PID: 3730 (mysqld)
Tasks: 28 (limit: 4560)
CGroup: /system.slice/mysql.service
└─3730 /usr/sbin/mysqld --daemonize --pid-file=/run/mysqld/mysqld.pid
окт 10 20:50:38 Zivko systemd[1]: Starting MySQL Community Server...
окт 10 20:50:38 Zivko systemd[1]: Started MySQL Community Server.
zivko@Zivko:~$
zivko@Zivko: ~
File Edit View Search Terminal Help
zivko@Zivko:~$ service mysql status
mysql.service - MySQL Community Server
Loaded: loaded (/lib/systemd/system/mysql.service; enabled; vendor preset: en
Active: inactive (dead) since Thu 2019-10-10 22:26:56 CEST; 3s ago
Main PID: 3730 (code=exited, status=0/SUCCESS)


zivko@Zivko: ~
File Edit View Search Terminal Help
zivko@Zivko:~$ sudo service mysql start
zivko@Zivko:~$
zivko@Zivko: ~
File Edit View Search Terminal Help
zivko@Zivko:~$ sudo apt install openssh-server
Reading package lists... Done
Building dependency tree
Reading state information... Done
zivko@Zivko: ~
File Edit View Search Terminal Help
zivko@Zivko:~$ sudo systemctl status ssh
ssh.service - OpenBSD Secure Shell server
Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enab
Active: active (running) since Thu 2019-10-10 22:42:16 CEST; 3min 2s ago
Main PID: 5460 (sshd)
Tasks: 1 (limit: 4560)
CGroup: /system.slice/ssh.service
└─5460 /usr/sbin/sshd -D
окт 10 22:42:16 Zivko systemd[1]: Starting OpenBSD Secure Shell server...
окт 10 22:42:16 Zivko sshd[5460]: Server listening on 0.0.0.0 port 22.
окт 10 22:42:16 Zivko sshd[5460]: Server listening on :: port 22.
окт 10 22:42:16 Zivko systemd[1]: Started OpenBSD Secure Shell server.
lines 1-12/12 (END)


zivko@Zivko: ~
File Edit View Search Terminal Help
zivko@Zivko:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 192.168.0.107 netmask 255.255.255.0 broadcast 192.168.0.255
inet6 fe80::bb62:a164:a7d5:8aa2 prefixlen 64 scopeid 0x20<link>
inet6 fec0::4ad4:9195:8704:8ce prefixlen 64 scopeid 0x40<site>
inet6 fec0::f963:f044:e762:d54c prefixlen 64 scopeid 0x40<site>
ether 00:15:5d:00:69:05 txqueuelen 1000 (Ethernet)
RX packets 52677 bytes 72705777 (72.7 MB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 27072 bytes 1964299 (1.9 MB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0x10<host>
loop txqueuelen 1000 (Local Loopback)
RX packets 472 bytes 40990 (40.9 KB)
RX errors 0 dropped 0 overruns 0 frame 0


192.168.0.107 - PuTTY
PuTTY Security Alert
The server's host key is not cached in the registry. You have no guarantee that the server is the computer you think it is.
The server's ssh-ed25519 key fingerprint is:
ssh-ed25519 255 9a:0a:93:66:2a:60:8f:50:aa:77:19:f0:42:cf:45:31
If you trust this host, hit Yes to add the key to PuTTY's cache and carry on connecting.
If you want to carry on connecting just once, without adding the key to the cache, hit No.
If you do not trust this host, hit Cancel to abandon the connection.
Yes
No
Cancel
Help
login as: zivko
zivko@192.168.0.107's password:


zivko@192.168.0.107's password:
Welcome to Ubuntu 18.04.3 LTS (GNU/Linux 5.0.0-23-generic x86_64)
* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/advantage
* Canonical Livepatch is available for installation.
- Reduce system reboots and improve kernel security. Activate at:
https://ubuntu.com/livepatch
150 packages can be updated.
88 updates are security updates.
Your Hardware Enablement Stack (HWE) is supported until April 2023.
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
zivko@Zivko:~$
zivko@Zivko:~$ ls -l
total 44
drwxr-xr-x 2 zivko zivko 4096 окт 10 14:21 Desktop
drwxr-xr-x 2 zivko zivko 4096 окт 10 14:21 Documents
drwxr-xr-x 2 zivko zivko 4096 окт 10 14:21 Downloads
-rw-r--r-- 1 zivko zivko 8980 окт 10 14:18 examples.desktop
drwxr-xr-x 2 zivko zivko 4096 окт 10 14:21 Music
drwxr-xr-x 2 zivko zivko 4096 окт 10 14:21 Pictures
drwxr-xr-x 2 zivko zivko 4096 окт 10 14:21 Public
drwxr-xr-x 2 zivko zivko 4096 окт 10 14:21 Templates
drwxr-xr-x 2 zivko zivko 4096 окт 10 14:21 Videos
zivko@Zivko:~$

```
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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> exit
Bye
zivko@Zivko:~$ mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 12
Server version: 5.7.27-0ubuntu0.18.04.1 (Ubuntu)

Copyright (c) 2000, 2019, Oracle and/or its affiliates. All rights reserved.

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

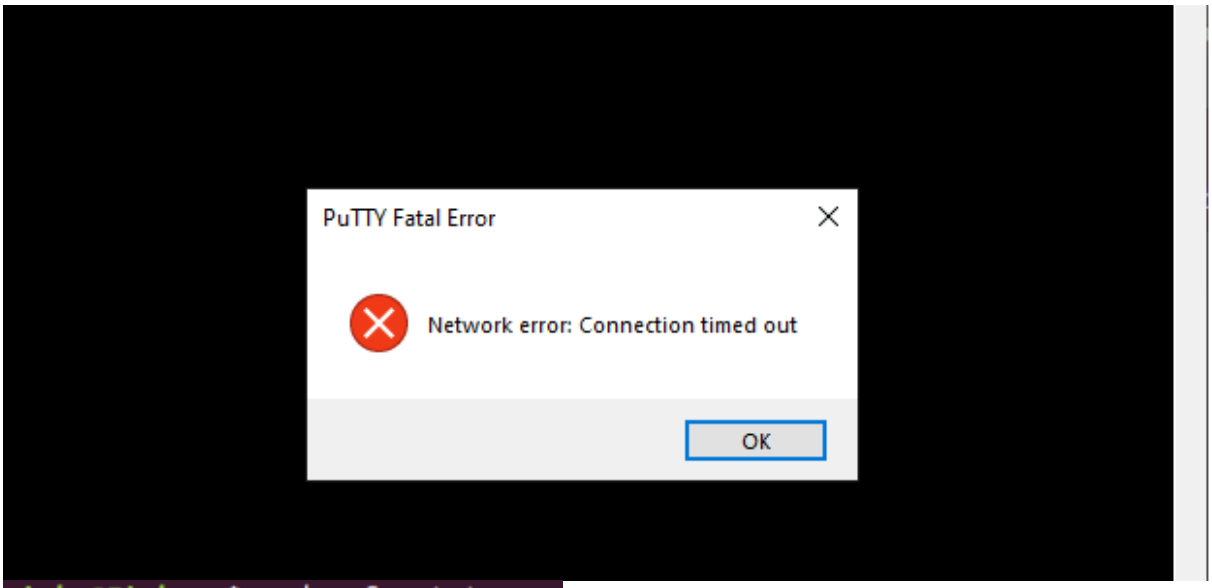
```
-----------+----------------------+------------------+---------------+
4 rows in set (0.00 sec)

mysql> Select user, authentication_string, plugin, host FROM mysql.user;
+------------------+-------------------------------------------+-----------------------+-----------+
| user             | authentication_string                     | plugin                | host      |
+------------------+-------------------------------------------+-----------------------+-----------+
| root             |                                           | auth_socket           | localhost |
| mysql.session    | *THISISNOTAVALIDPASSWORDTHATCANBEUSEDHERE | mysql_native_password | localhost |
| mysql.sys        | *THISISNOTAVALIDPASSWORDTHATCANBEUSEDHERE | mysql_native_password | localhost |
| debian-sys-maint | *E07C34E911498D16E050335A7CA3ED7C0210976B | mysql_native_password | localhost |
+------------------+-------------------------------------------+-----------------------+-----------+
4 rows in set (0.00 sec)

mysql>
mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT user, authentication_string, plugin, host FROM mysql.user;
+------------------+-------------------------------------------+-----------------------+-----------+
| user             | authentication_string                     | plugin                | host      |
+------------------+-------------------------------------------+-----------------------+-----------+
| root             | *0D580D7435F0247ECB684C098A6725B645FE62BC | mysql_native_password | localhost |
| mysql.session    | *THISISNOTAVALIDPASSWORDTHATCANBEUSEDHERE | mysql_native_password | localhost |
| mysql.sys        | *THISISNOTAVALIDPASSWORDTHATCANBEUSEDHERE | mysql_native_password | localhost |
| debian-sys-maint | *E07C34E911498D16E050335A7CA3ED7C0210976B | mysql_native_password | localhost |
+------------------+-------------------------------------------+-----------------------+-----------+
4 rows in set (0.00 sec)

mysql>
```




```
zivko@Zivko:~$ sudo ufw status
Status: active
zivko@Zivko:~$
```

```
zivko@Zivko:~$ sudo ufw disable
Firewall stopped and disabled on system startup
zivko@Zivko:~$
```

```
zivko@Zivko:~$ sudo ufw allow 22
Rule added
Rule added (v6)
zivko@Zivko:~$
```

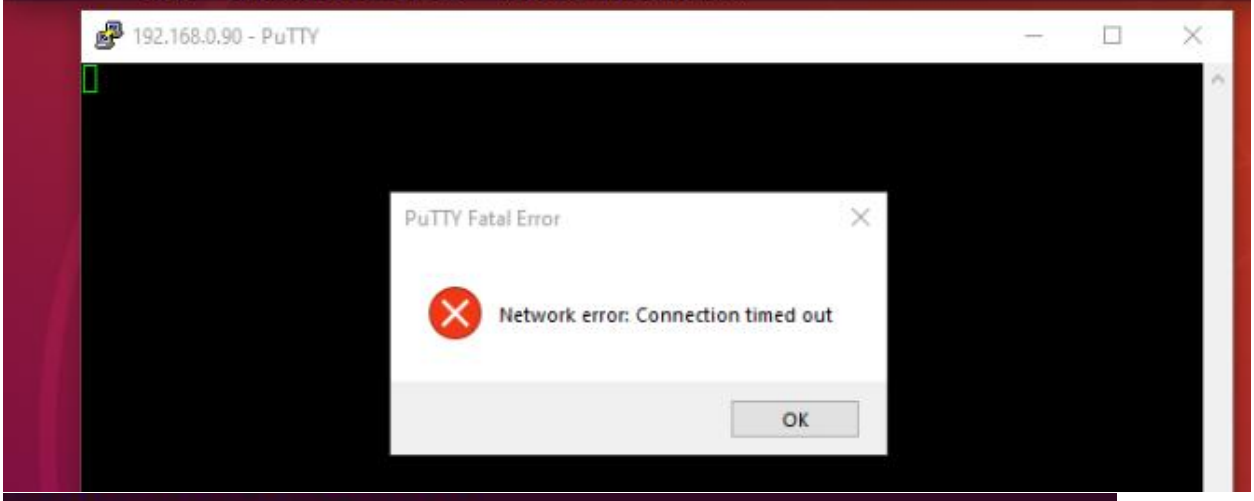
```
zivko@Zivko:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22                         ALLOW       Anywhere
22 (v6)                    ALLOW       Anywhere (v6)

zivko@Zivko:~$
```

```
zivko@Zivko:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.0.98  netmask 255.255.255.0  broadcast 192.168.0.255
        inet6 fe80::8501:4cf5:1a7f:e687  prefixlen 64  scopeid 0x20<link>
        ether 00:15:5d:00:69:04  txqueuelen 1000  (Ethernet)
        RX packets 72818  bytes 17454830 (17.4 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 3443  bytes 1050265 (1.0 MB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
```




```
# Allow incoming TCP ports
TCP_IN = "20,21,22,25,53,80,110,143,443,465,587,993,995,5001"

# Allow outgoing TCP ports
TCP_OUT = "20,21,22,25,37,53,80,110,113,443,587,993,995"

# Allow incoming UDP ports
UDP_IN = "20,21,53,80,443"

# Allow outgoing UDP ports
# To allow outgoing traceroute add 33434:33523 to this list
UDP_OUT = "20,21,37,53,113,123"
```

```
# Allow incoming TCP ports
TCP_IN = "20,21,22,25,53,80,110,143,443,465,587,993,995,5001,3306"

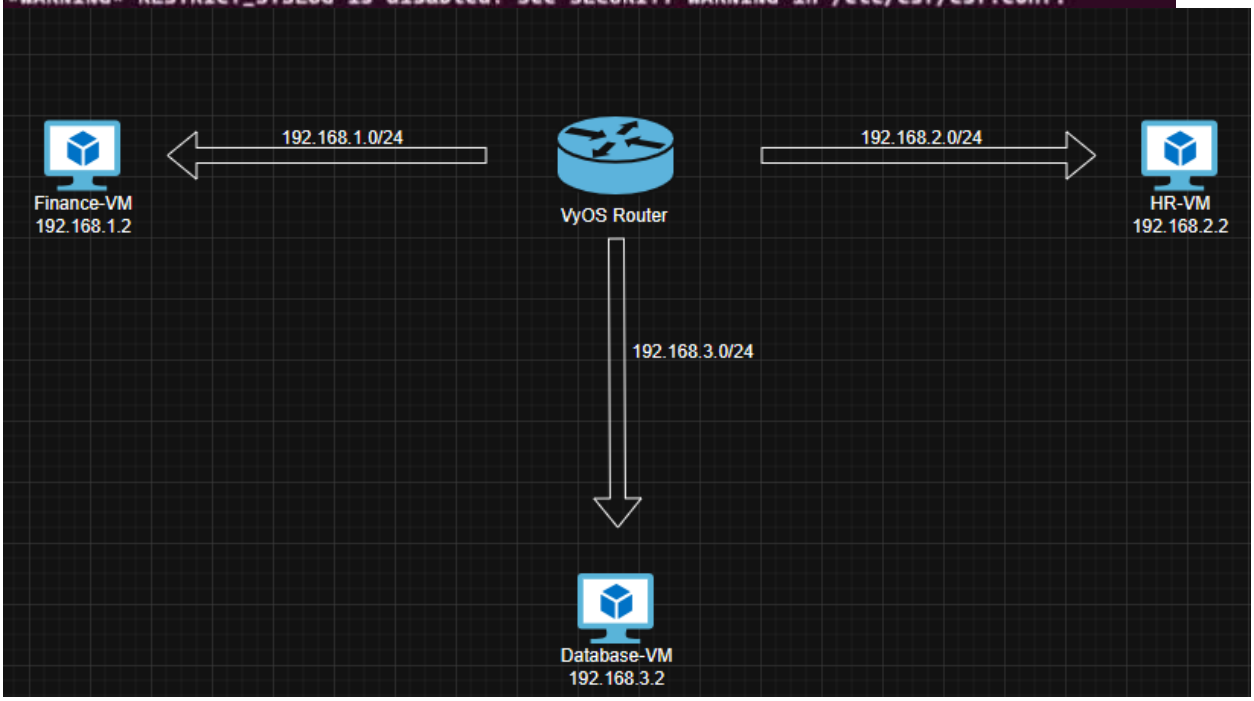
# Allow outgoing TCP ports
TCP_OUT = "20,21,22,25,37,53,80,110,113,443,587,993,995,3306"

# Allow incoming UDP ports
UDP_IN = "20,21,53,80,443,3306"

# Allow outgoing UDP ports
# To allow outgoing traceroute add 33434:33523 to this list
UDP_OUT = "20,21,37,53,113,123,3306"
ALLOWOUT  all opt -- in * out !lo  0.0.0.0/0  -> 0.0.0.0/0
ALLOWIN  all opt -- in !lo out *  0.0.0.0/0  -> 0.0.0.0/0
DENYOUT  all opt    in * out !lo  ::/0  -> ::/0
DENYIN  all opt    in !lo out *  ::/0  -> ::/0
ALLOWOUT  all opt    in * out !lo  ::/0  -> ::/0
ALLOWIN  all opt    in !lo out *  ::/0  -> ::/0
csf: FASTSTART loading Packet Filter (IPv4)
csf: FASTSTART loading Packet Filter (IPv6)
DROP  all opt -- in * out *  0.0.0.0/0  -> 0.0.0.0/0
INVALID  tcp opt -- in !lo out *  0.0.0.0/0  -> 0.0.0.0/0
INVALID  tcp opt -- in * out !lo  0.0.0.0/0  -> 0.0.0.0/0
DROP  all opt    in * out *  ::/0  -> ::/0
INVALID  tcp opt    in !lo out *  ::/0  -> ::/0
INVALID  tcp opt    in * out !lo  ::/0  -> ::/0
csf: FASTSTART loading csf.deny (IPv4)
csf: FASTSTART loading csf.allow (IPv4)
ACCEPT  icmp opt -- in !lo out *  0.0.0.0/0  -> 0.0.0.0/0   icmptype 8 limit: avg 1/sec burst 5
LOGDROPIN  icmp opt -- in !lo out *  0.0.0.0/0  -> 0.0.0.0/0   icmptype 8
ACCEPT  icmp opt -- in !lo out *  0.0.0.0/0  -> 0.0.0.0/0
ACCEPT  icmp opt -- in * out !lo  0.0.0.0/0  -> 0.0.0.0/0
ACCEPT  ipv6-icmp opt    in !lo out *  ::/0  -> ::/0
ACCEPT  ipv6-icmp opt    in * out !lo  ::/0  -> ::/0
ACCEPT  all opt -- in !lo out *  0.0.0.0/0  -> 0.0.0.0/0   ctstate RELATED,ESTABLISHED
ACCEPT  all opt -- in * out !lo  0.0.0.0/0  -> 0.0.0.0/0   ctstate RELATED,ESTABLISHED
ACCEPT  all opt    in !lo out *  ::/0  -> ::/0   ctstate RELATED,ESTABLISHED
ACCEPT  all opt    in * out !lo  ::/0  -> ::/0   ctstate RELATED,ESTABLISHED
csf: FASTSTART loading TCP_IN (IPv4)
csf: FASTSTART loading TCP6_IN (IPv6)
csf: FASTSTART loading TCP_OUT (IPv4)
csf: FASTSTART loading TCP6_OUT (IPv6)
csf: FASTSTART loading UDP_IN (IPv4)
csf: FASTSTART loading UDP6_IN (IPv6)
csf: FASTSTART loading UDP_OUT (IPv4)
csf: FASTSTART loading UDP6_OUT (IPv6)
```

```
ACCEPT  all opt -- in lo out *  0.0.0.0/0  -> 0.0.0.0/0
ACCEPT  all opt -- in * out lo  0.0.0.0/0  -> 0.0.0.0/0
LOGDROPOUT  all opt -- in * out !lo  0.0.0.0/0  -> 0.0.0.0/0
LOGDROPIN  all opt -- in !lo out *  0.0.0.0/0  -> 0.0.0.0/0
ACCEPT  all opt    in lo out *  ::/0  -> ::/0
ACCEPT  all opt    in * out lo  ::/0  -> ::/0
LOGDROPOUT  all opt    in * out !lo  ::/0  -> ::/0
LOGDROPIN  all opt    in !lo out *  ::/0  -> ::/0
csf: FASTSTART loading DNS (IPv4)
csf: FASTSTART loading DNS (IPv6)
LOCALOUTPUT  all opt -- in * out !lo  0.0.0.0/0  -> 0.0.0.0/0
LOCALINPUT  all opt -- in !lo out *  0.0.0.0/0  -> 0.0.0.0/0
LOCALOUTPUT  all opt    in * out !lo  ::/0  -> ::/0
LOCALINPUT  all opt    in !lo out *  ::/0  -> ::/0

*WARNING* RESTRICT_SYSLOG is disabled. See SECURITY WARNING in /etc/csf/csf.conf.
```



Secure Network Architecture Report

Name: Fortune Migwi Wanyoike
Date: February 24, 2026
Role: Junior Cybersecurity Engineer

Network Segmentation Goals and Access Controls

Purpose of this Network Configuration

The purpose of this configuration is to secure sensitive healthcare data by isolating the Database Server into a dedicated subnet (192.168.3.0/24) and enforcing strict access control via a VyOS

router. This transition from a flat network to a segmented one ensures that only authorized systems—specifically the Finance department—can interact with the database.

- **Why is network segmentation necessary for securing sensitive systems?** It reduces the "blast radius" of a security breach. If the HR subnet is compromised, the attacker cannot automatically move laterally to the database because the VyOS router acts as a gatekeeper.
- **What specific access controls were implemented to ensure data security?** We implemented Layer 4 stateful firewall rules that allow only TCP traffic on port 3306 (MySQL) from the Finance subnet while explicitly dropping and logging all attempts from the HR subnet.
- **How does this configuration align with security best practices?** This configuration aligns with the Principle of Least Privilege (PoLP) and CIS benchmarks by restricting network paths to the absolute minimum necessary for business operations.

Access Control Rules and Justification

- **What traffic was explicitly allowed?** Incoming TCP traffic to the Database Server (192.168.3.2) on port 3306 originating from the Finance Subnet (192.168.1.0/24).
- **What traffic was denied?** All traffic originating from the HR Subnet (192.168.2.0/24) destined for the Database Subnet (192.168.3.0/24).
- **Why were these rules necessary for security?** Finance requires database access for operational efficiency (e.g., payroll processing), whereas HR has no business requirement to reach the MySQL service, making their access a high-security risk.

Firewall Rules and Network Traffic Control

Firewall Configuration and Implementation

The firewall rules were implemented on the VyOS router to serve as the central enforcement point. Each rule was designed to monitor traffic as it enters the database interface from external subnets.

- **Subnets involved:** Finance (192.168.1.0/24), HR (192.168.2.0/24), and Database Server (192.168.3.0/24).
- **Specific firewall rules:** A "Permit" rule for port 3306 was applied for Finance, while a "Drop" rule with logging enabled was applied for the HR subnet.
- **Justification:** These rules ensure that financial records remain confidential and are only accessible by validated systems within the Finance department.

Traffic Validation and Analysis

Network Connectivity Tests

- **Finance-VM:** Successfully established connectivity to the Database VM. Ping tests were successful, and the MySQL client successfully authenticated with the server.
- **HR-VM:** Connection attempts failed. Ping tests returned a timeout, and MySQL connection attempts were blocked by the VyOS firewall.
- **Troubleshooting:** We identified that static routes were initially missing; adding return routes in the Netplan configuration for both the Finance-VM and DB-VM resolved the communication gap.

Capturing and Analyzing Traffic

- **Explain the role of tcpdump in analyzing network traffic:** `tcpdump` allows for real-time packet inspection to verify that firewall rules are being processed correctly and that traffic is originating from the expected source IPs.
- **Summarize key traffic patterns observed:** Authorized traffic showed a standard TCP three-way handshake followed by MySQL protocol data from the Finance IP (192.168.1.2).
- **Provide examples of expected vs. unexpected traffic behavior:** We observed successful handshakes from the Finance subnet. Unexpectedly, no packets were visible from the HR IP at the DB-VM level, proving the VyOS router silently dropped the packets as intended.

Reflection on Network Security Implementation

Effectiveness of Network Segmentation

The segmentation was highly effective. By placing the database in its own subnet, we successfully enforced a "walled garden" approach where only a single authorized path exists for sensitive data retrieval.

Challenges and Adjustments

- **What challenges were encountered?** The primary challenge was ensuring bidirectional communication. Even when the firewall allowed traffic in, the DB-VM did not initially know the path back to the Finance subnet.
- **How were they resolved?** This was resolved by implementing persistent static routes in the Netplan YAML files on both the client and server VMs.

Future Security Enhancements

- **What improvements could be made?** Implement an Intrusion Detection System (IDS) to scan the allowed Finance traffic for application-layer attacks like SQL injection.

- **How could logging be integrated?** Integrate VyOS firewall logs into a centralized Syslog server or SIEM for real-time alerts on unauthorized access attempts from restricted subnets like HR.

Broader Security Implications

- **Why is network segmentation critical?** It limits the "blast radius" of a breach, ensuring a compromised workstation in one department cannot easily access the entire enterprise data center.
- **How do controlled access points improve resilience?** They provide centralized enforcement and visibility, making it easier for security teams to monitor and block malicious activity.
- **What real-world attacks does this mitigate?** Segmentation is a primary defense against ransomware propagation and unauthorized lateral movement by internal or external actors.